
Projects

Echoes in the Mists (Souls-Like)

2024 - 2025

- Built scalable, data-driven systems and tools supporting 30+ unique enemies and 8+ unique gameplay-altering charms.
- Optimized performance for a single large-level environment housing 100+ enemies
- Profiled & resolved bottlenecks using tools such as the Unity Profiler and Frame Debugger.
- Collaborated on a multidisciplinary team coordinating engineering efforts with design, art, and narrative.

Mosaic: A Universal Gameplay Framework

2022 - Present

- Engineered a modular, data-driven package for Unity, inspired by Unreal's Gameplay Ability System, enforcing cross-compatible features and making abilities/systems reusable across players, NPCs, and bosses.
- Accelerated prototyping and production by reducing boilerplate and enabling rapid feature iteration for developers and designers.
- Adopted in production for a public demo and presented at GDC 2025.

Experience

Adjunct Professor | West Virginia University

2025 - Present

- Developed and taught graduate-level UI/UX emphasizing technical implementation in development.
- Partnered with AbleGamers to embed accessibility as a technical requirement in production pipelines.

Teaching Assistant | Data Structures and Algorithms 2 – RIT

2024

- Conducted code reviews of C++/OpenGL data structure implementations, providing feedback to students.
- Mentored students, reinforcing understanding of data structures and programming concepts.

Programmer | Faculty Directed Development – SUNY Poly

2023

- Developed C++ features and debugging tools to support teaching Data Structures & Algorithms.

Organizer | Game Design Club – SUNY Poly

2022 - 2023

- Co-led the college's first student-led initiative to send 12 students to GDC, securing **\$50K+ funding** through cross-department and state partnerships.

Education

MS, Game Design and Development
Rochester Institute of Technology

BS, Interactive Media and Game Design
SUNY Polytechnic Institute

- Minor: **Computer Science** (advanced coursework)
- GPA 3.96 | Summa Cum Laude
- Academic Achievement Award (highest GPA)

AAS, Computer Information Systems
Hudson Valley Community College

Skills

Languages:

C++, C#, Assembly, HLSL, GLSL

Engines & Tech:

Unity, Unreal, Godot, EnTT/ECS, Git (project standards), GitHub, Jira

Systems & Performance:

ECS, profiling & optimization, physics simulation, PS5 Dev Kit

Data & Integrations:

Firebase(data-tracking), SQL, JSON(save systems)